

Multi Agentic RAG-Based Travel Planning & Optimization System

Leveraging LLMs · CrewAI Multi-Agent Framework · Real-Time Flight & Hotel APIs for Intelligent Trip Planning

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01

Literature Survey

Research Papers · Comparative Analysis · Research Gap

5 Marks

01

An Agent-Based RAG Architecture for Intelligent Tourism Assistance: The Valencia Case Study

Bonetti et al. (2025) · DOI: 10.3390/tourhosp6050266 · <https://doi.org/10.3390/tourhosp6050266>

Problem Addressed	Architecture / Model	System Capabilities	Limitations
- Fragmented travel info causes decision paralysis - Cognitive Load Theory: high mental switching cost - Static RAG lacks geospatial & temporal awareness - No unified conversational travel interface	- Multi-agent ReAct RAG system 3 Agents: Retrieval, Events, Geospatial - LLM: Mistral Small 3.1 - ChromaDB - LangChain ReAct Orchestration	- Unified conversational interface - Geospatial awareness - Multi-agent coordination - Constraint-based trip	- City-specific (Valencia, Spain) - Static dataset, no real-time APIs - High inference latency - GPT-4o needed for preprocessing - No flight/hotel booking flow

ADDITIONAL RESEARCH PAPERS — 2 & 3

02

RAG and Agentic Assistant: A Combined Approach

Moez Ben Hajhmida1(B) and Edward A. Lee2 - springer (2025)

Technique: Naive/Advanced/Modular RAG; hybrid retrieval; chunking strategies; embedding models

Our contribution: We integrate live APIs (SerpAPI, Apify) into the RAG-style pipeline

 No real-time API integration; no travel-domain multi-tool agentic use

03

TravelAgent: An AI Assistant for Personalized Travel Planning

Aili Chen, Xuyang Ge, Ziquan Fu♥, Yanghua Xiao, Jiangjie Chen - arxiv (2024)

Technique: ReAct paradigm: chain-of-thought reasoning interleaved with tool-based action execution

Our contribution: We apply ReAct via CrewAI with 4 specialized travel agents

 Evaluated on knowledge benchmarks; no travel-specific multi-API orchestration

COMPARATIVE ANALYSIS OF RELATED WORKS

Paper	LLM Used	RAG	Agents	Live APIs	Flight/Hotel	Itinerary	Evaluation
Valencia RAG (Bonetti 2025)	Mistral 3.1	✓ Hybrid	✓ 3 Agents	✗	✗	✗	RAGAS 0.897
RAG Survey (Gao 2024)	Multiple	✓ Advanced	Partial	✗	✗	✗	Survey
ReAct (Yao 2023)	PaLM/GPT	✗	✓ ReAct	Search API	✗	✗	HotpotQA
CrewAI (Moura 2024)	Any LLM	Supported	✓ Roles	Tool-based	✗	✗	Framework
OUR PROJECT (2025–26)	Groq Llama 3.3	✓ + APIs	✓ 4 Agents	✓ Live	✓ Yes	✓ Yes	Not yet decided

RESEARCH GAP IDENTIFIED & OUR CONTRIBUTION

X Static knowledge bases only (no live data)



✓ Datasets + Live SerpAPI for flights + Apify for hotels — always current

X City-specific, single-language (Spanish only)



✓ Global destinations, English, user-defined origin + destination

X No end-to-end trip booking workflow



✓ Full pipeline: Search → Select → Itinerary → Download PDF

X No role-specific travel agent architecture



✓ 4 CrewAI agents: Flight · Hotel · Itinerary · Recommendation

X High-cost LLM inference (GPT-4o required)



✓ Groq API (Llama 3.3-70B): free tier

02

Base Paper Explanation

Architecture · Algorithms · Evaluation Metrics

5 Marks

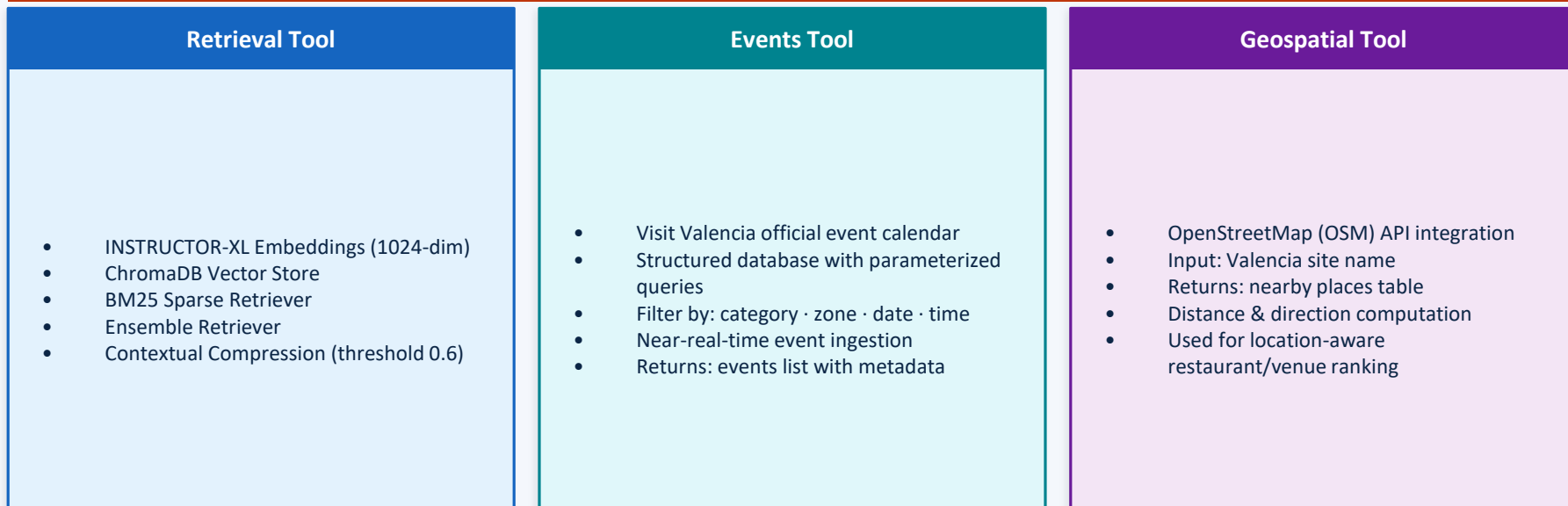
02

BASE PAPER: SYSTEM ARCHITECTURE & REACT FLOW

ReAct Agent Pipeline — Valencia Tourism RAG



3 Specialized Tools — Separation of Concerns Architecture



BASE PAPER: ALGORITHMS, METRICS & LIMITATIONS vs OUR IMPROVEMENTS

ALGORITHMS USED		LIMITATIONS → OUR IMPROVEMENTS	
1	INSTRUCTOR-XL Task-specific 1024-dim embeddings without fine-tuning	✗ Valencia only (Spanish)	✓ Global destinations, English
2	BM25 (sparse) Best Match 25 keyword ranking — weight 0.4 in ensemble	✗ Static dataset, no live data	✓ Dataset + Live SerpAPI + Apify per request
3	ChromaDB (dense) Semantic vector search — weight 0.6 in ensemble	✗ Sequential latency in ReAct	✓ Groq LLM inference speed
4	Ensemble Retriever Hybrid BM25 + dense for holistic relevance	✗ GPT-4o preprocessing cost	✓ No external preprocessing needed
5	Contextual Compression Filters chunks below similarity threshold 0.6	✗ No trip booking flow	✓ Full itinerary + PDF export
6	ReAct Loop Reason → Act → Observe → Iterate until final answer	✗ No user preference handling	✓ Special instructions input field

03

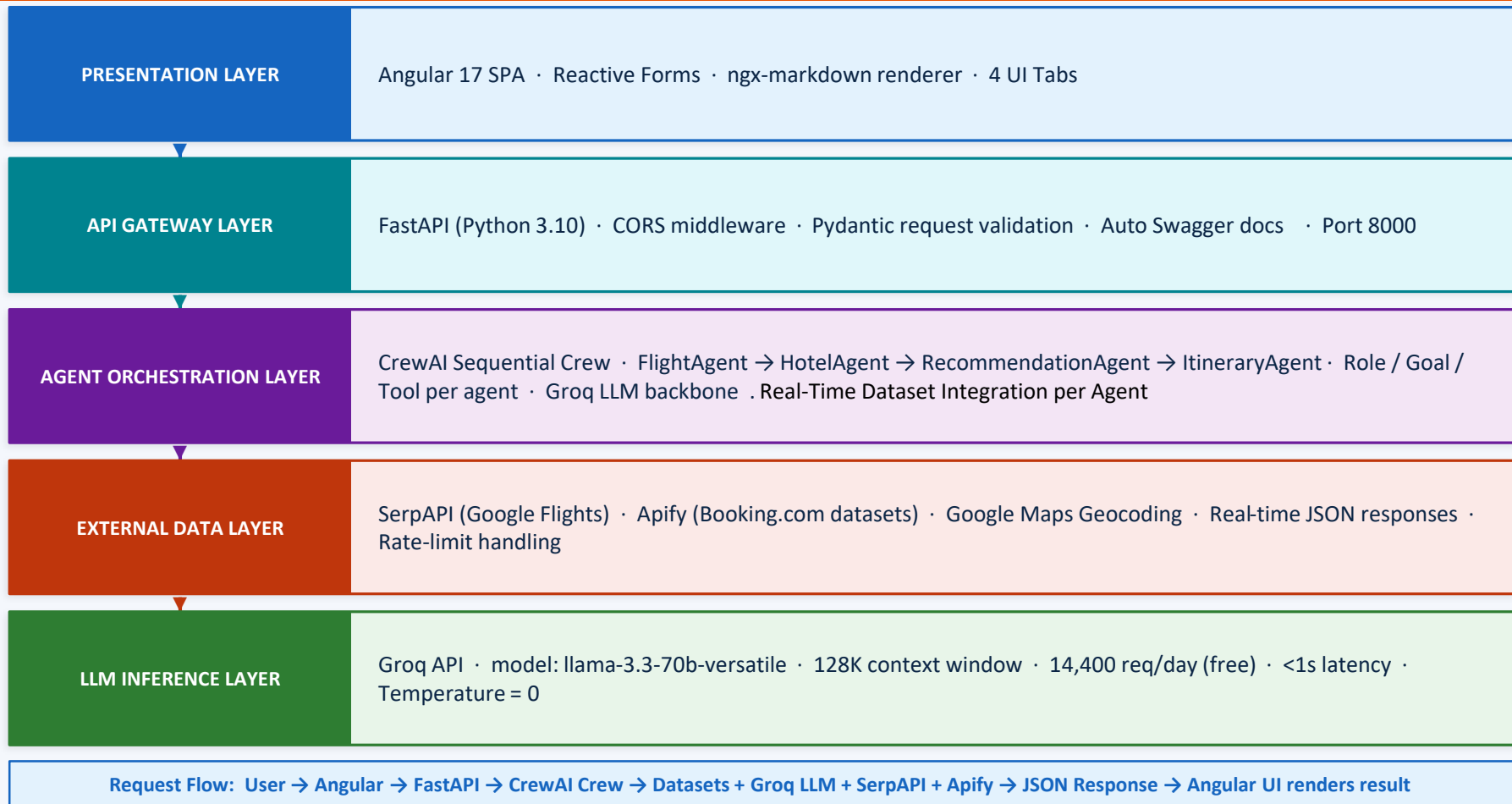
Technical Architecture

System Design · Agents · Data Flow · Modules

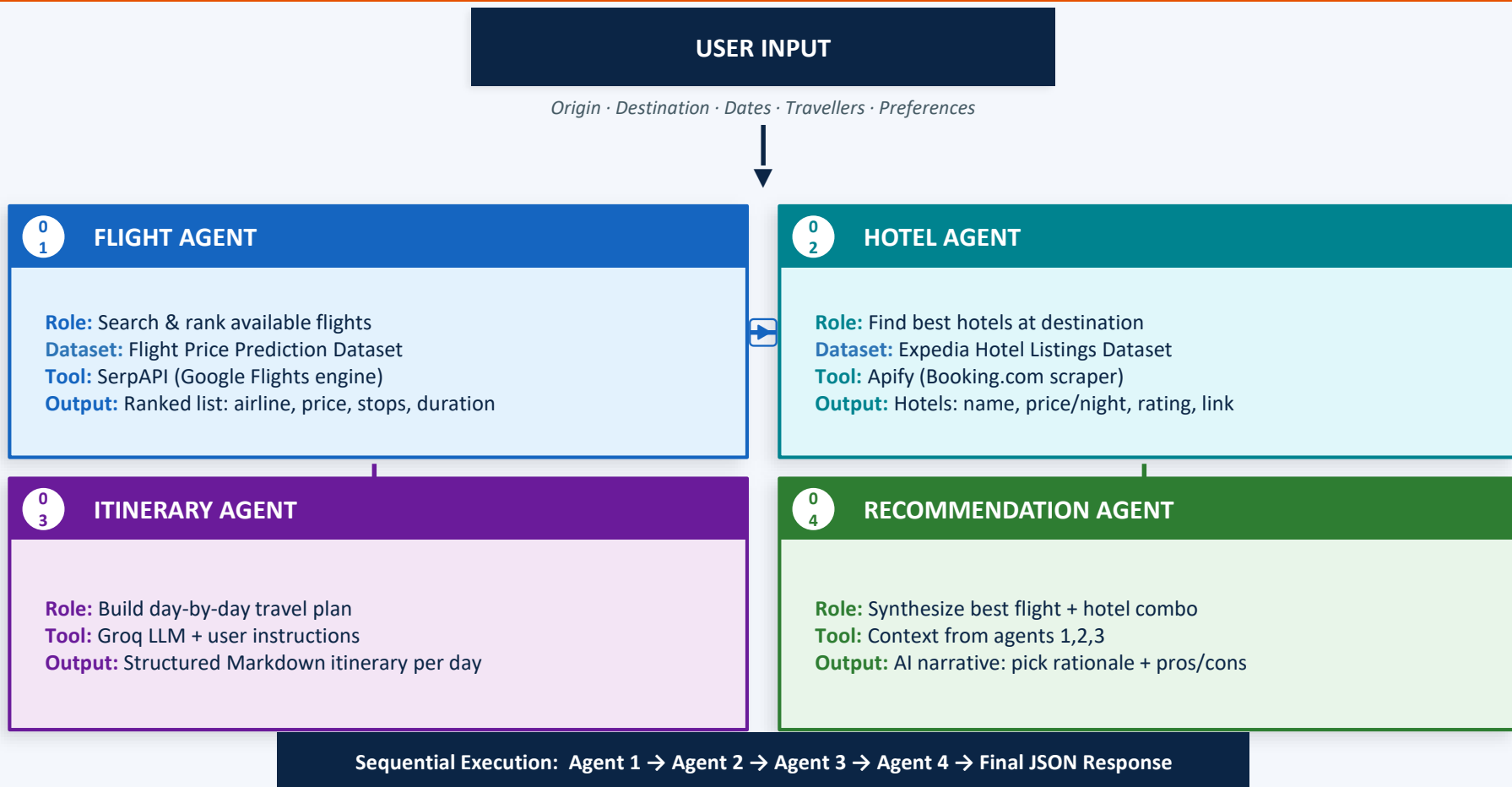
5 Marks

03

PROPOSED SYSTEM ARCHITECTURE — 5-LAYER DESIGN



CREWAI MULTI-AGENT WORKFLOW — 4 SPECIALIZED AGENTS



04

Dataset Strategy

Kaggle Datasets · Live APIs · Preprocessing · Ethics

5 Marks

04

DATASET STRATEGY — KAGGLE DATASETS FOR TRAINING & EVALUATION



Kaggle Datasets used for pre-training context, model evaluation, and prompt grounding

01

Expedia Hotel Listings Dataset

CSV/Tabular

~500K hotel records

Use: Train hotel ranking logic · price/rating preprocessing · feature extraction for HotelAgent prompts



kaggle.com/datasets/expedia-group/hotel-listing-data

Cols: hotel_id, city, country, rating, price_per_night, amenities, review_count, latitude, longitude

02

Flight Price Prediction Dataset

CSV/Tabular

~300K flight records

Use: Understand flight pricing patterns · inform FlightAgent ranking criteria · validate SerpAPI outputs



kaggle.com/datasets/shubhambathwal/flight-price-prediction

Cols: airline, source_city, departure_time, stops, arrival_time, destination_city, duration, price

03

TripAdvisor Hotel Reviews Dataset

Text/NLP

~20K reviews

Use: Fine-tune sentiment understanding for recommendation agent · extract preference keywords from reviews



kaggle.com/datasets/andrewmvd/trip-advisor-hotel-reviews

Cols: Review (text), Rating (1–5) — used for NLP-based hotel preference modelling

04

World Cities Travel Itinerary Dataset

Geospatial/Text

~40K cities

Use: Validate destination names · build city-to-country lookup for ItineraryAgent context injection



kaggle.com/datasets/viswanathanc/world-cities-dataset

Cols: city, country, population, latitude, longitude, timezone — used for geo-grounding prompts

DATA SOURCES — LIVE APIs + PREPROCESSING PIPELINE

SerpAPI (Flight Data)

Google Flights engine
Real-time: price, airline
stops, duration, dep_id
JSON structured response

Apify (Hotel Data)

Booking.com scraper
Hotel name, price/night
rating, images, booking link
5 free actor runs/month

Groq LLM (Itinerary)

Llama 3.3-70B model
Prompt-driven generation
No static corpus needed
<1s inference latency

Kaggle (Grounding)

4 datasets for context
Hotel, Flight, Reviews
World cities lookup
Prompt enrichment

PREPROCESSING PIPELINE (5 STEPS)

1

API Call

SerpAPI / Apify
fetched per user request

2

JSON Parse

Extract: price, airline
rating, hotel name, link

3

Filter & Sort

Remove nulls · sort by
price, rating, duration

4

Prompt Build

Structured context
injected into agent prompt

5

LLM Generate

Groq processes context
→ Markdown output

05

Methodology & Tools

Tech Stack · Implementation Strategy · Deployment

5 Marks

10 Marks

TECHNOLOGY STACK — TOOL SELECTION & JUSTIFICATION

Python 3.10

Core Language

Strong AI/ML ecosystem, clean syntax for agent logic, and reliable package management.

CrewAI

Agent Framework

Designed for structured multi-agent workflows with role-based task execution and context passing.

Groq (Llama 3.3)

LLM Provider

Fast inference, 128K context window, free-tier access, and stable large-model performance.

FastAPI

Backend Server

High performance, async support, built-in validation (Pydantic), auto API docs, and CORS handling.

Angular 17

Frontend

SPA architecture, TypeScript safety, reactive forms, and clean UI component structure.

SerpAPI

Flight Data

Structured real-time flight data with pricing and route details.

Apify

Hotel Data

Real-time hotel listings including pricing, ratings, and availability.

GitHub

Version Control

Version tracking, branch management, and secure project collaboration.

IMPLEMENTATION STRATEGY & DEPLOYMENT PLAN

IMPLEMENTATION PHASES



Phase 1 — Environment Setup

Created Python virtual environment and installed dependencies.



Phase 2 — FastAPI Backend

Built backend using **FastAPI**.
Defined CrewAI agents (Flight, Hotel, Itinerary, Recommendation) with clear roles and tools



Phase 3 — API Integration

Integrated live datasets : Google Flights real-time fares, Booking.com hotel listings



Phase 4 — LLM Integration

Connected **Groq** (Llama 3.3 70B model).



Phase 5 — Angular Frontend

4 tabs: Flights / Hotels / AI Picks / Trip Plan · Markdown render · PDF export



Phase 6 — Evaluation (Pending)

Planned qualitative testing on travel queries

DEPLOYMENT PLAN

Local Development (Current)

Backend → Port 8000
Frontend → Port 4200
Access via browser (localhost)

Backend — Production

Docker container + uvicorn ASGI
Cloud hosting (AWS EC2 / Railway)

Frontend — Deployment

static hosting
API routing configuration for backend communication

Error Handling & Reliability

Input validation using Pydantic models
Fallback responses when external APIs fail

Scalability Plan

Load balancer for FastAPI instances
Groq rate-limit fallback handler

THANK

YOU !

